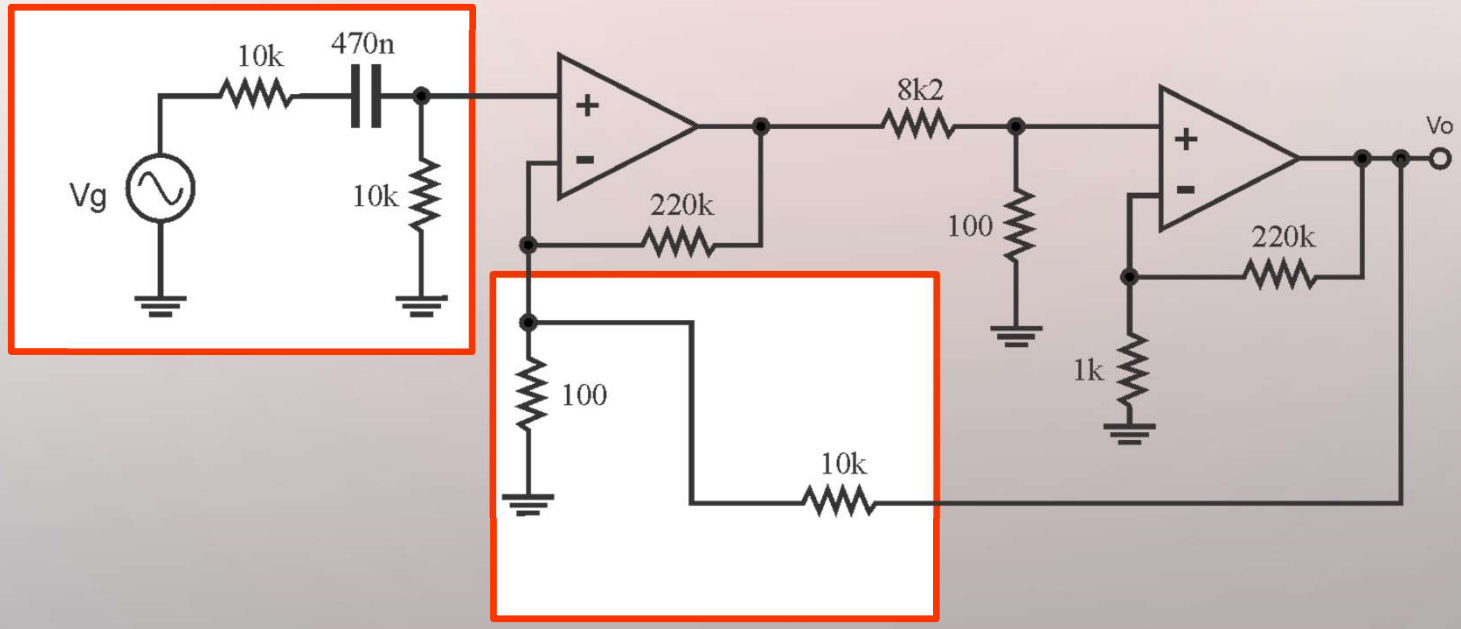
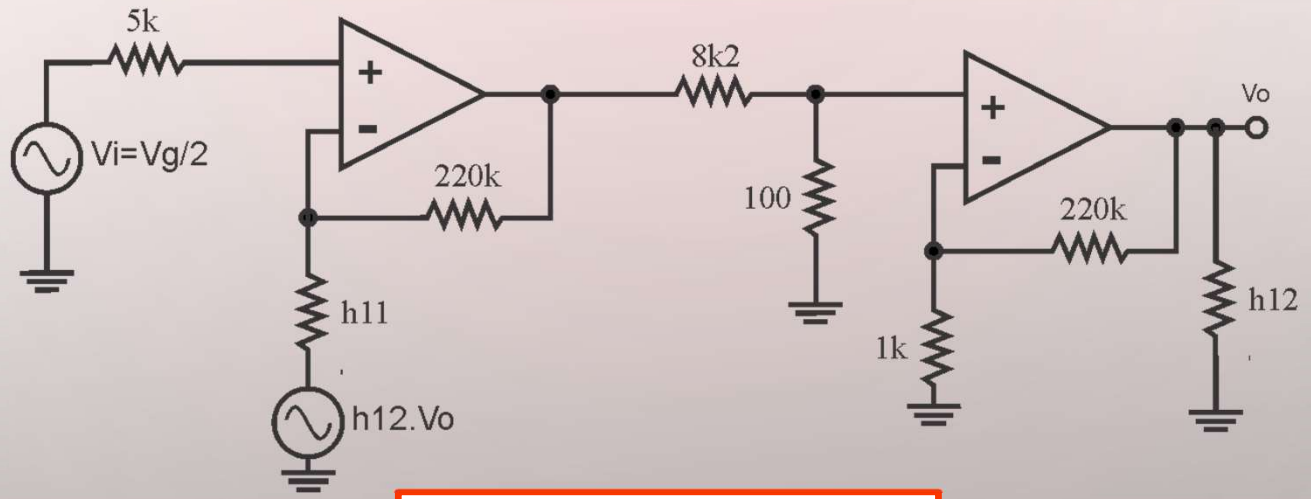
LABORATORIO N°1

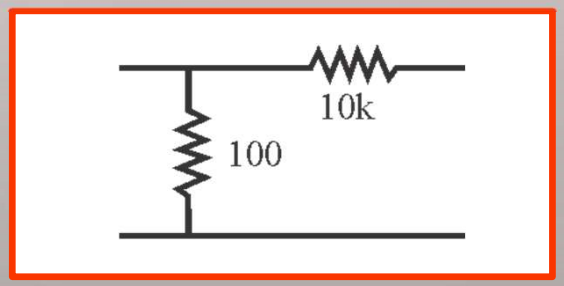
ESTABILIDAD DE LOS AMPLIFICADORES REALIMENTADOS

AMPLIFICADOR REALIMENTADO



AMPLIFICADOR REALIMENTADO ADAPTADO





AMPLIFICADOR DE UN POLO REALIMENTADO

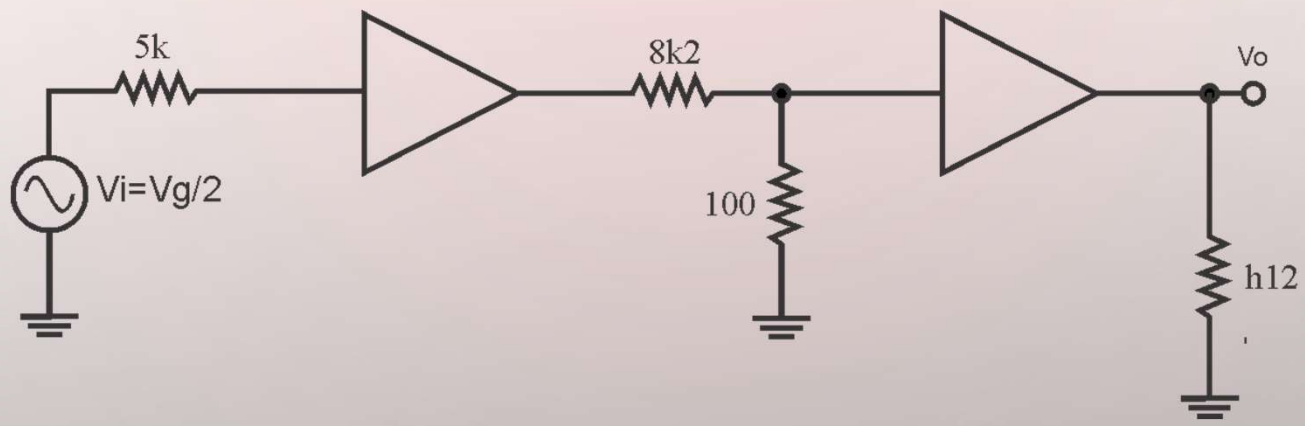
$$a(jf) = \frac{a_0}{1 + \frac{jf}{f_0}}$$

$$A(jf) = \frac{a(jf)}{1 + a(jf) * f} = \frac{A_0}{1 + \frac{jf}{f_o'}}$$

$$f_0' = f_0 * (1 + a_0 * f)$$

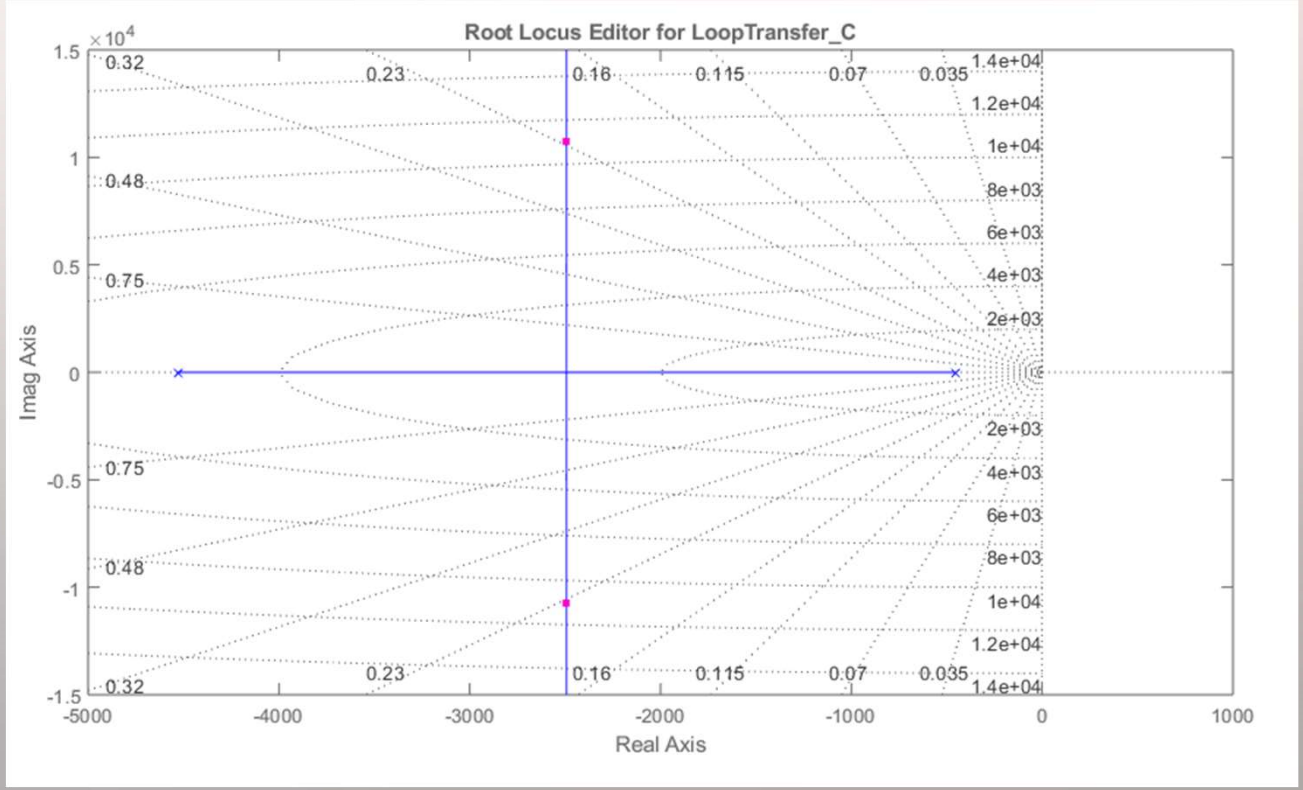
$$A_0 = \frac{a_0}{1 + a_0 * f}$$

GANANCIA DIRECTA ($f = 0$)



$$a_{v_{f=0}} = \frac{5857}{\left(1 + \frac{jf}{455Hz}\right)\left(1 + \frac{jf}{4{,}53kHz}\right)}$$

ESTABILIDAD – LUGAR DE RAÍCES

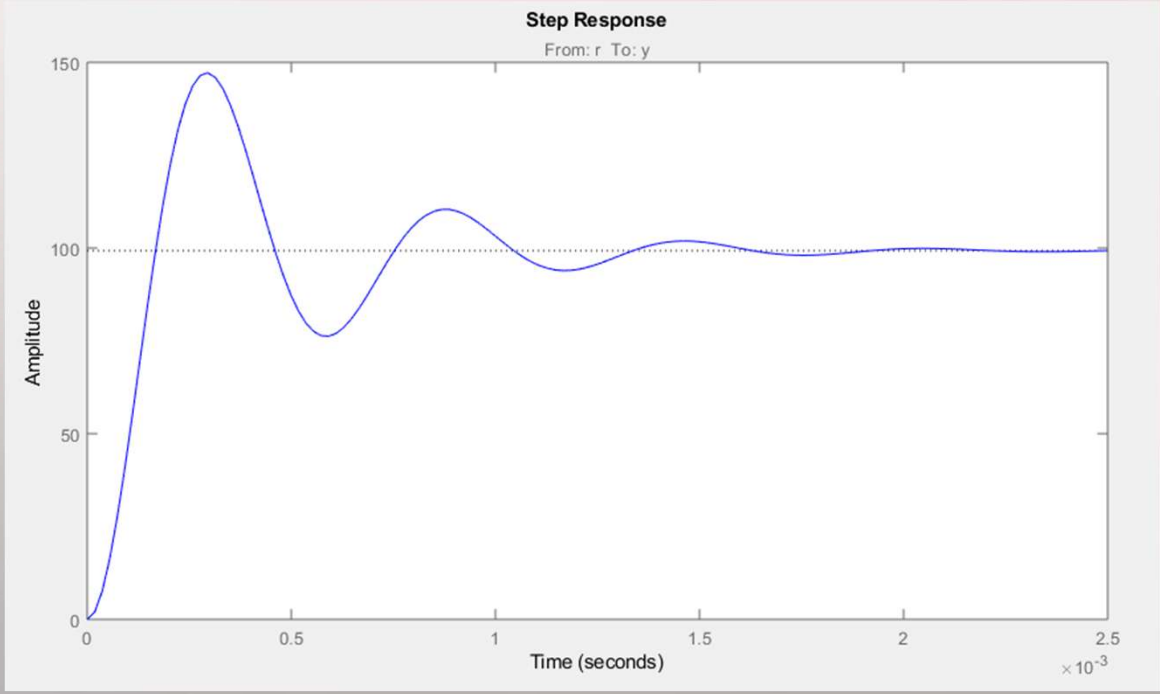


$$f_0 = \sqrt{fp_1 * fp_2(1 + T_0)}$$

$$Q = \frac{f_0}{fp_1 + fp_2}$$

$$Si\ Q > \frac{1}{2} \begin{cases} \alpha = \dfrac{f_0}{2Q} \\ \beta = \alpha\sqrt{4Q^2 - 1} \end{cases}$$

ESTABILIDAD – RESPUESTA AL ESCALÓN

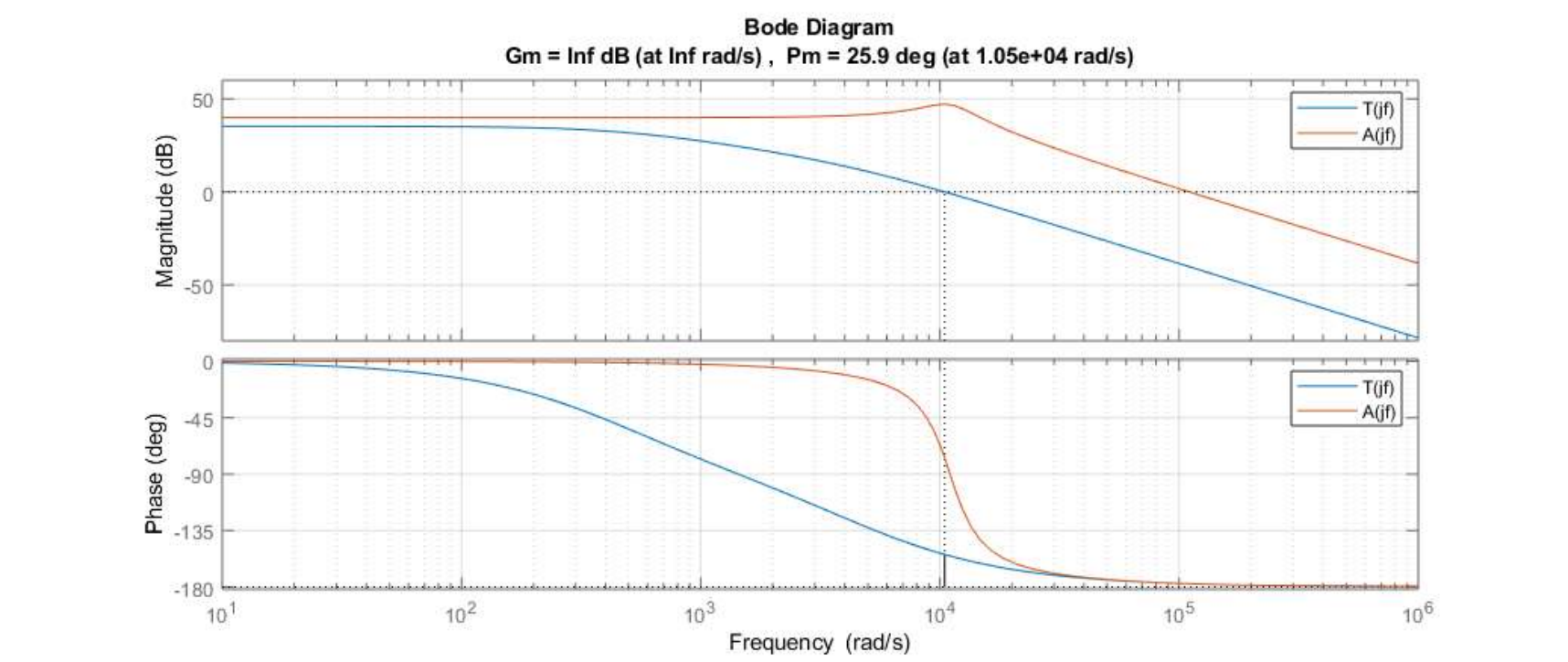


Frecuencia de oscilación: $1/\beta$

Coeficiente de atenuación: α

Sobreimpulso: $\mathrm{SI} = \mathrm{e}^{-\frac{\pi}{\sqrt{4Q^2-1}}}$

ESTABILIDAD - BODE



Sobrepico: $f_x = f_0\sqrt{1 - \frac{1}{2Q^2}}$